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RESEARCH ARTICLE

A Stacking Ensemble Framework Leveraging Synthetic Data for Accurate and Stable Crop Yield Forecasting

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ABSTRACT With the rapid increase in world's population and changing climate patterns, accurate crop yield forecasting is essential to ensure food security and sustainable agriculture. This study presents a yield prediction framework consisting of Stacking Ensemble Model (SEM) and its Optimized variant (OSEM), which integrates real-world agricultural data with synthetic data generated using the Prophet time-series model. The ensemble comprises Random Forest, XGBoost, Decision Tree, and K-Nearest Neighbors as base learners, with an Extra Trees Regressor as the meta-learner. OSEM incorporates hyperparameter tuning and dimensionality reduction using Singular Value Decomposition to enhance predictive performance and reduce feature noise. Extensive experiments on globally distributed and region-specific crop datasets demonstrate that OSEM consistently outperforms both SEM and state-of-the-art baselines, achieving $R^2 = 0.996$ and MAE = 0.185 t/ha. Stability and convergence analysis, supported by Wilcoxon and Friedman tests, further confirm the robustness and reliability of the proposed model for scalable crop yield forecasting.

INDEX TERMS Crop yield prediction, stacking ensemble learning, synthetic data augmentation, Prophet, XGBoost, random forest, decision tree, KNN, extra tree, SVD.

I. INTRODUCTION

Agriculture plays a pivotal role in ensuring global food security and economic stability, especially in developing nations [1], [2]. With the global population projected to reach 9.7 billion by 2050 [3], increasing crop productivity is critical to meet future food demands. Crops like wheat, maize, and soybeans are not only essential to the global diet but also serve as major exports [4], [5], [6]. Accurate crop yield forecasting enables governments and organizations to make informed decisions regarding resource planning, trade policy, and climate adaptation strategies [7].

Traditional yield estimation methods such as surveys, expert assessments, or trend-based models, often lack timely updates and adaptability, especially in the face of rapidly changing climate conditions [8], [9]. This has fueled the shift towards data-driven approaches leveraging Machine Learning (ML) and Artificial Intelligence (AI) to uncover complex, non-linear relationships in crop, soil, and environmental data [10]. ML models, such as Gradient Boosting, SVM, and Random Forest, have outperformed classical regression techniques in recent applications [11], [12], while also assisting in yield demand forecasting and spatial planning [13].

The convergence of ML with remote sensing and agronomy has demonstrated promising results in yield estimation across diverse geographic and climatic zones. Recent studies underscore the utility of combining temporal and spatial features, historical records, and multisource observations to enhance prediction robustness [14], [15], [16], [17]. However, despite progress, challenges persist in data quality, regional variability, and model interpretability.

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Although significant advancements have been made in ML-based crop yield forecasting, almost none of the studies have holistically integrated synthetic data, ensemble learning, and optimization in a unified framework. Moreover, there remains a lack of generalizable models that perform robustly across both actual and simulated future data scenarios, especially when accounting for regional diversity. This study addresses these challenges by proposing a hybrid framework that synthesizes past trends, leverages model diversity, and improves adaptability through optimization. To assist readers unfamiliar with the terminology used throughout the study, a complete list of abbreviations is provided in Table 1.

A. STUDY OBJECTIVES

This study explores gaps in the existing literature and aims to address the following questions:

- How can synthetic data generated using time-series models like Prophet enhance traditional crop yield prediction systems?
- What is the performance impact of a stacking ensemble model that integrates synthetic data compared to individual machine learning models?
- How can dimensionality reduction and hyperparameter tuning improve the robustness and efficiency of yield prediction models?

B. RESEARCH CONTRIBUTIONS

The major contributions of this study are as follows:

- We propose a stacking ensemble framework (SEM and OSEM) that integrates multiple base learners with dimensionality reduction and hyperparameter tuning to improve crop yield forecasting accuracy.
- Synthetic data generated via Prophet model is used exclusively for validation and testing, enabling realistic generalization to unseen and future-like agricultural scenarios.
- The final ensemble integrates XGBoost, Random Forest, Decision Tree, and KNN as base learners, with an Extra Trees Regressor as the meta-learner to effectively capture and leverage inter-model dependencies.
- Dimensionality reduction using Truncated SVD is applied to retain 80% of the most informative variance, helping to reduce overfitting and computational load.
- We benchmark our proposed model against both classical machine learning models and recent transformer-based architectures (TabNet, TabPFN), showing consistent superiority in performance across key metrics.
- Robustness and reliability of the proposed model are assessed through convergence and stability analysis across 10-folds, along with statistical validation using the Wilcoxon signed-rank test and the Friedman test.
- The model is evaluated using k-fold cross-validation and region-specific yield forecasting scenarios, demonstrating strong generalization and scalability for diverse agricultural settings.

TABLE 1. List of abbreviations used in this study.

Abbreviation	Full Form
ML	Machine Learning
AI	Artificial Intelligence
DL	Deep Learning
SEM	Stacking Ensemble Model
OSEM	Optimized Stacking Ensemble Model
CYP	Crop Yield Prediction
XGB	Extreme Gradient Boosting
RF	Random Forest
DT	Decision Tree
KNN	K-Nearest Neighbors
ETR	Extra Trees Regressor
BR	Bagging Regressor
GB	Gradient Boosting
LR	Linear Regression
SVM	Support Vector Machine
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
1D-CNN	One-Dimensional Convolutional Neural Network
3D-CNN	Three-Dimensional Convolutional Neural Network
LSTM	Long Short-Term Memory
ConvLSTM	Convolutional Long Short-Term Memory
SVR	Support Vector Regression
MLR	Multivariate Linear Regression
GPR	Gaussian Process Regression
MH-SA	Multihead Self-Attention
ViT	Vision Transformer
NAS	Neural Architecture Search
DNNR	Deep Neural Network Regressor
ARSE	Absolute Relative Squared Error
LASSO	Least Absolute Shrinkage and Selection Operator
CMLR	Constrained Multivariate Linear Regression
CDT	Causal Decision Tree
CKNN	Constrained K-Nearest Neighbors
WOFOST	World Food Studies Simulation Model
MCYFS	Monitoring Agricultural Resources Crop Yield Forecasting System
SVD	Singular Value Decomposition
TabNet	Attentive Interpretable Tabular Learning
TabPFN	Tabular Prior-data Fitted Network
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
MSE	Mean Square Error
MAPE	Mean Absolute Percentage Error
R^2	Coefficient of Determination
CV	Cross Validation
AAV	Autonomous Aerial Vehicle
NDVI	Normalized Difference Vegetation Index
FAO	Food and Agriculture Organization
API	Application Programming Interface
USDA	United States Department of Agriculture
t/ha	Tons per Hectare
hg/ha	Hectograms per Hectare
kg/ha	Kilograms per Hectare
Std.	Standard Deviation
KS	Kolmogorov–Smirnov

The structure of this study is organized as follows: Section II provides relevant related work followed by Section III, which outlines the methodology of the proposed SEM and OSEM models. After that, Section IV covers data processing, synthetic data, exploratory data analysis, and with that Section V presents the experimental setup with detailed evaluations and stability analysis. Furthermore, Section VI offers a brief discussion and comparison with

TABLE 2. Comparison of existing deep learning and machine learning models with limitations.

Author(s)	Year	Models	Study Area	Predicted Crops	Limitations
Deep Learning Models					
Shook et al. [26]	2021	LSTM	USA, Canada	Soybean	Lack of genomic data limits genotype-specific predictions
Oikonomidis et al. [25]	2022	CNN, LSTM, DNN	Global	Maize, Wheat, Soybean	Limited training data causes overfitting
Ramzan et al. [24]	2023	DNN	Pakistan	Tea	Ground truth yield data is sparse and collected via surveys
Rama Devi et al. [18]	2023	CNN, CNN-LSTM	India	Rice	Small dataset with dependency on weather and soil features
Valarezo-Plaza et al. [20]	2024	1D-CNN	Canada	Canola	Excludes soil/genotype data and only uses HSI-based features
Olisah et al. [19]	2024	DNNR	Nigeria	Corn	Uses only variables easily accessible to small farmers
MirhoseiniNejad et al. [21]	2024	ConvLSTM-ViT	USA	Corn, Soybean	Training dependent on multi-source data which may be unavailable in some locations
Miranda et al. [37]	2024	PI-LSTM	Switzerland	Cereal	Does not generalize to other crop types or global regions
Zhang et al. [23]	2024	Transformer, CNN, CNN-LSTM	China	Wheat	Drone data collection may not scale for large regions
Najjar et al. [22]	2025	LSTM, DT, RF	Uruguay, Germany, Argentina	Maize, Soybean	Focuses on subfield-level predictions with limited generalizability
Machine Learning Models					
Paudel et al. [35]	2021	GB, RF, SVR, KNN	Multiple European countries	Wheat, Barley, Sunflower, Sugar beet, Potatoes	Cannot predict for large-scale datasets
Sharifi [36]	2021	Gaussian Process Regression, DT, KNN	Iran	Barley	Unable to predict when exceeding certain yield value
Cedric et al. [34]	2022	DT, LR, KNN	West African countries	Rice, Maize, Cassava, Seed cotton, Yams, Bananas	Data quality was not good or sparse
Morales & Villalobos [33]	2023	RF, ANN, SVR	Spain	Sunflower, Wheat	Inavailability of large amount of datasets
Shafi et al. [31]	2023	RF, XGB, LASSO	Pakistan	Wheat	Heavily relied on UAV-based dataset
Sharma et al. [29]	2023	DT, RF, XGB, CNN, LSTM	India	Various crops (Rice, Maize, Wheat, etc.)	Less effective where data is sparse or unreliable
Yadav et al. [28]	2024	Bagging Regressor, KNN, DT, GB, RF, LR	USA	Various crops including wheat, maize, etc.	Rely on accurate and complete dataset
Ashfaq et al. [32]	2024	RF, SVM, LASSO	Pakistan	Wheat	Remote sensing data can be inconsistent
Hoque et al. [27]	2024	GB, KNN, Multivariate LR	India	Rice, Maize, Cassava, Seed Cotton, Yams, Bananas	Dependent on accuracy of climatic and pesticides data
Tamayo-Vera et al. [30]	2025	RF, GB	Canada	Potatoes	Used only historical data for prediction

state-of-the-art models, along with limitations and finally, Section VII concludes the study and suggests potential directions for future research.

II. LITERATURE REVIEW

To establish a comprehensive understanding of existing approaches in crop yield forecasting, this section reviews recent advancements in both machine learning and deep learning techniques, highlighting their methodologies, strengths, and limitations in agricultural prediction tasks.

A. DEEP LEARNING MODELS FOR CYP

Recent advancements in deep learning (DL) have significantly improved the performance of crop yield forecasting by leveraging complex, non-linear patterns from multisource datasets. Rama Devi et al. [18] combined hyperspectral imaging with CNN and optimized CNN-LSTM to monitor crop growth stages and predict yield, achieving $R^2 = 0.893$, RMSE = 0.13, and MAE = 0.14. Similarly, Olisah et al. [19] employed a Deep Neural Network Regressor (DNNR) using soil and weather variables to forecast corn yield in Nigeria, achieving MAE = 0.0172 t/ha with a novel error metric (ARSE). Valarezo-Plaza et al. [20] introduced an edge-deployable, optimized 1D-CNN model for canola yield prediction using drone imagery, improving efficiency by reducing model size by 92% with $R^2 = 0.772$. These studies show the growing relevance of lightweight and interpretable DL architectures for both accuracy and real-world deployment in agricultural settings.

DL models integrating spatial, temporal, and attention mechanisms are gaining momentum in capturing agro-climatic variations. MirhoseiniNejad et al. [21] proposed a hybrid ConvLSTM-3D-CNN-ViT model to predict soybean yield using multispectral data, achieving high correlation and low RMSE. Najjar et al. [22] emphasized explainability in yield forecasting by applying LSTM models with feature attribution methods across satellite imagery of soybean, wheat, and rapeseed. Zhang et al. [23] compared LSTM, stacking ensemble, and LSTM-MH-SA (Multihead Self-Attention), concluding that LSTM-MH-SA achieved the highest accuracy with a 29.58% improvement and $R^2 = 0.81$. These findings support the integration of temporal modeling and attention mechanisms for more interpretable and reliable crop yield predictions.

Furthermore, hybrid DL models have demonstrated success in combining remote sensing, weather, and other data modalities for yield prediction. Ramzan et al. [24] utilized NDVI features and meteorological data in a NAS-optimized DNN architecture to predict tea yield in Pakistan, achieving $R^2 = 0.99$ and MAE = 2.26 Kg/ha. The study by Oikonomidis et al. [25] evaluated hybrid combinations including CNN-DNN and CNN-LSTM for soybean prediction using over 25,000 samples and found CNN-DNN to be the most effective ($R^2 = 0.87$, RMSE = 0.266). Shook et al. [26] proposed an LSTM-based model combining genotype and weather data for soybean forecasting, achieving $R^2 = 0.796$, demonstrating the value of genetic-environmental integration. A comparative summary of these recent deep learning models, including architectures, domains, and limitations is provided in Table 2.

Despite these advancements, many existing DL approaches face limitations such as restricted generalizability, dependency on sparse or local datasets, or lack of integration across modalities. In contrast, our proposed hybrid framework combines spatial-temporal modeling with synthetic data augmentation and robust interpretability, offering a scalable and generalized solution for crop yield prediction across heterogeneous agro-climatic zones.

B. MACHINE LEARNING MODELS FOR CYP

Recent machine learning (ML) studies in crop yield forecasting emphasize ensemble learning, remote sensing integration, and regional data modeling. Hoque et al. [27] used pesticide, climate, and yield data across six Indian crops to train Gradient Boosting (GB), KNN, and Multivariate Linear Regression (MLR), where GB achieved the highest accuracy (99.99%, MAE = 0.0182 t/ha). Similarly, Yadav et al. [28] applied an ensemble framework to the USDA dataset (2003-2013), where Bagging Regressor and Random Forests (RF) exceeded 98% accuracy. Sharma et al. [29] evaluated both classical and deep models—RF achieved 98.96%, while CNN led DL models with the lowest loss (0.00060). Tamayo-Vera et al. [30] demonstrated the benefits of localized prediction using 35 years of data in Canada, achieving $R^2 = 0.99$ with RF. These studies underscore the growing shift toward combining historical, climatic, and image-based data in regional ML pipelines.

Temporal and multi-source forecasting is another active area. Shafi et al. [31] analyzed remote sensing and sowing date variation to optimize wheat yield predictions, with LASSO performing best during early-season stages ($R^2 = 0.93$). Ashfaq et al. [32] combined climate, satellite, soil, and socioeconomic data (2017-2022) for yield prediction in Pakistan, where RF achieved 97% accuracy. Morales and Villalobos [33] took a different approach by generating synthetic data through crop simulations for five Spanish regions, and found RF models reduced RMSE to 35% surpassing traditional baselines (42%). These works validate the benefits of integrating remote sensing and synthetic augmentation into ML pipelines, while also revealing gaps in addressing real-time adaptability and large-scale deployment.

Other studies highlight cross-regional generalization and system design. Cedric et al. [34] built a decision-support system for six crops across West Africa using CMLR, CDT, and CKNN; the latter achieved the best performance ($R^2 = 0.95$, MAE = 0.160 Kg/ha). Paudel et al. [35] proposed a modular yield prediction system using WOFOST climate features and ML models, surpassing the MCYFS benchmark. Sharifi [36] applied Gaussian Process Regression to Sentinel-2 and climate data for barley prediction (2015-2019), reaching $R^2 = 0.84$. A comparison of these ML models, along with their limitations and performance trade-offs, is summarized in Table 2.

While these ML models offer strong regional performance, several suffer from data sparsity, limited scalability, or dependence on high-quality inputs. In contrast, our proposed synthetic-ensemble framework combines robust feature selection, model stacking, and synthetic data generalization to address these limitations and ensure adaptability across diverse agricultural zones.

III. PROPOSED METHODOLOGY

This section introduces the proposed models, SEM and OSEM, developed for crop yield forecasting. Subsection III-A provides an overview of the SEM and OSEM frameworks, while Subsection III-B details the architectural design of the proposed models. Lastly, Subsection III-C presents a step-by-step explanation of the algorithmic implementation.

A. OVERVIEW OF SEM & OSEM

This subsection introduces the overall design of the proposed crop yield prediction framework based on two ensemble models: the Stacking Ensemble Model (SEM) and the Optimized Stacking Ensemble Model (OSEM). As illustrated in Figure 1, the proposed framework begins with historical data collection (1990-2013) and synthetic data generation (2014-2024) from weather, crop, and pesticide datasets. After preprocessing, SEM combines four base learners Random Forest (RF), Decision Tree (DT), XGBoost (XGB), and K-Nearest Neighbors (KNN), whose outputs are passed to a meta-learner, Extra Trees Regressor (ETR). This

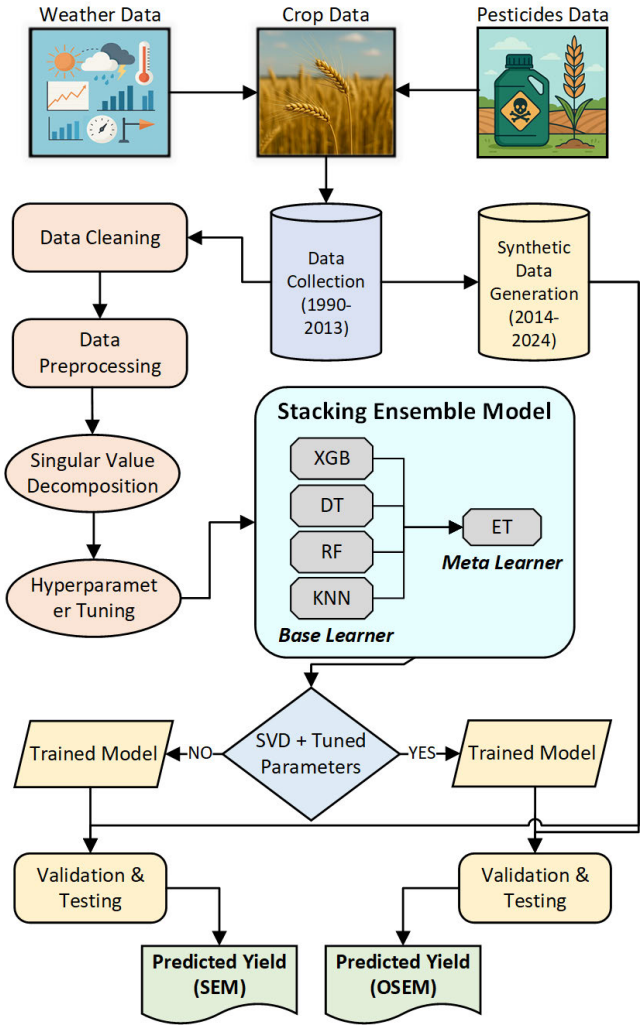


FIGURE 1. Overview pipeline of the proposed SEM & OSEM framework.

layered structure enables the model to learn complementary patterns and supports validation of both standard (SEM) and optimized (OSEM) configurations.

In SEM, each base learner is independently trained on the same dataset. Their individual predictions are then stacked and used as input for the meta-learner to generate the final output. This two-level ensemble framework (Level-0 base models and Level-1 meta-learner) enables the model to exploit complementary learning strengths and reduce bias.

To enhance generalization, we introduce synthetic data generated using the Prophet time-series forecasting model. Prophet was selected due to its ability to model long-term trends and seasonality efficiently, especially in agricultural data. This synthetic data simulates future climate and yield patterns, making the model more robust against distributional shifts.

Building upon SEM, we present OSEM by integrating two additional optimization steps: (i) grid search-based hyperparameter tuning for both base and meta-learners, and (ii) dimensionality reduction using Singular Value Decomposition (SVD), which retains 80% of the most informative components. These refinements help reduce model complexity and improve generalization.

Both models are validated using 10-fold cross-validation to ensure robustness and prevent overfitting. The comprehensive workflow, shown in Figure 2, outlines the end-to-end SEM/OSEM process, including data preprocessing, feature engineering, synthetic data generation, model training, and performance evaluation.

B. PROPOSED MODEL ARCHITECTURE

This subsection explains the architecture and data flow of the proposed crop yield prediction framework, as illustrated in Figure 3. The architecture is structured into five color-coded layers, each representing a major phase of the model pipeline:

Layer 1- Data Collection (Blue): Historical datasets related to crop yield, rainfall, temperature, and pesticide usage are collected from global sources and unified into a single structured dataset.

Layer 2- Data Preprocessing (Yellow): This stage includes handling missing values, feature renaming, identification of categorical and numerical variables, and definition of input/output mappings. Duplicate records are removed to ensure data integrity.

Layer 3- Feature Engineering and Synthetic Data Generation (Green): Prophet is used to generate synthetic yield, temperature, rainfall, and pesticide data spanning the next 11 years. These synthetic datasets are appended to actual data to enrich training and testing diversity. Categorical variables are encoded using one-hot encoding, and continuous features are normalized using StandardScaler. Dimensionality reduction is applied using SVD to retain the most informative 80% of features and reduce noise.

Layer 4- Model Prediction (Pink): This layer implements the core SEM and OSEM frameworks. The base learners (XGB, RF, DT, KNN) are trained in parallel and their outputs are passed to the meta-learner (ETR). For OSEM, grid search optimization is applied to select the best hyperparameters, and SVD-reduced features are used to enhance training efficiency.

Layer 5- Evaluation and Output (Gray): Final predictions are evaluated using standard regression metrics: MAE, MSE, RMSE, MAPE, and R^2. The results provide insights into both model accuracy and generalization capability on unseen synthetic data.

This layered architecture ensures that the proposed models can adapt to evolving agro-climatic trends and heterogeneous regional datasets, contributing to scalable and future-ready crop yield forecasting solutions.

C. PROPOSED ALGORITHMS AND MODEL COMPONENTS

This subsection presents both the individual components of the proposed SEM and its enhanced version, OSEM. They consist of multiple base learners at Level 0 (XGBoost, Random Forest, Decision Tree, and KNN), and a meta learner at Level 1 (Extra Trees Regressor). Each model contributes

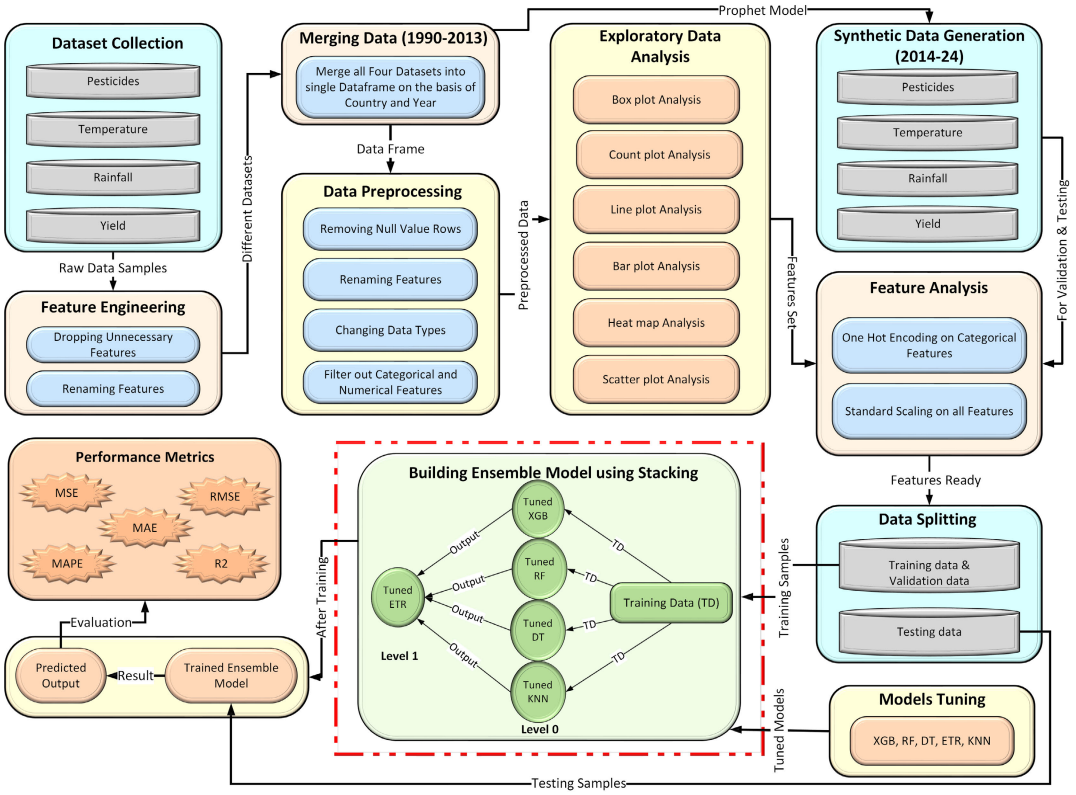


FIGURE 2. Comprehensive block diagram of SEM/OSEM framework from data collection to model evaluation.

uniquely to capture nonlinearities, feature relevance, and robustness in crop yield forecasting.

1) BASE LEARNERS IN SEM/OSEM

- **XGBoost (XGB)** is a gradient boosting framework known for its regularization and handling of sparse data [38].
- **Random Forest (RF)** is a bagging-based ensemble of decision trees that captures variance through randomized sampling [39].
- **Decision Tree (DT)** builds interpretable models through recursive feature splits based on minimum variance [40].
- **K-Nearest Neighbors (KNN)** estimates output based on proximity to feature-wise neighbors, suitable for locally sensitive data [41]. However, recent studies suggest that t-SNE may outperform KNN in certain unsupervised contexts due to its superior clustering and trend identification capabilities [42], [43].

2) META LEARNER

- **Extra Trees Regressor (ETR)** is an ensemble method that differs from RF by using full random splits, improving model de-correlation [44].

3) STACKING ENSEMBLE MODEL (SEM)

The SEM uses all base learners to generate predictions which are then fed into the meta-learner. Its algorithmic flow is given in Algorithm 1.

Algorithm 1 Stacking Ensemble Model (SEM)

Input: Training data X_{train}, Y_{train}
Output: Final prediction $\hat{y}_{\text{final}}$
Initialize Base Learners $f_1, f_2, \ldots, f_n$ (e.g., XGBoost, Random Forest, Decision Tree, KNN)
for each base learner f_i **do**
    Train base model f_i on X_{train}, Y_{train}
    Predict crop yield $\hat{y}_i = f_i(X_{\text{train}})$
end for
Stack predictions from all base models: $\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n$
Initialize meta-learner M (Extra Trees Regressor)
Train meta-learner M on the stacked predictions: $\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n$
Final prediction $\hat{y}_{\text{final}} = M(\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n)$
Return: $\hat{y}_{\text{final}}$

4) OPTIMIZED STACKING ENSEMBLE MODEL (OSEM)

OSEM further enhances SEM by applying hyperparameter tuning and dimensionality reduction using Singular Value Decomposition (SVD). The process is outlined in Algorithm 2.

IV. DATA PROCESSING AND EXPLORATORY ANALYSIS

This section discusses in detail the data, its processing and analysis using graphical implementations. The subsection IV-A presents the details about the data and its collection, followed by the preprocessing of the data

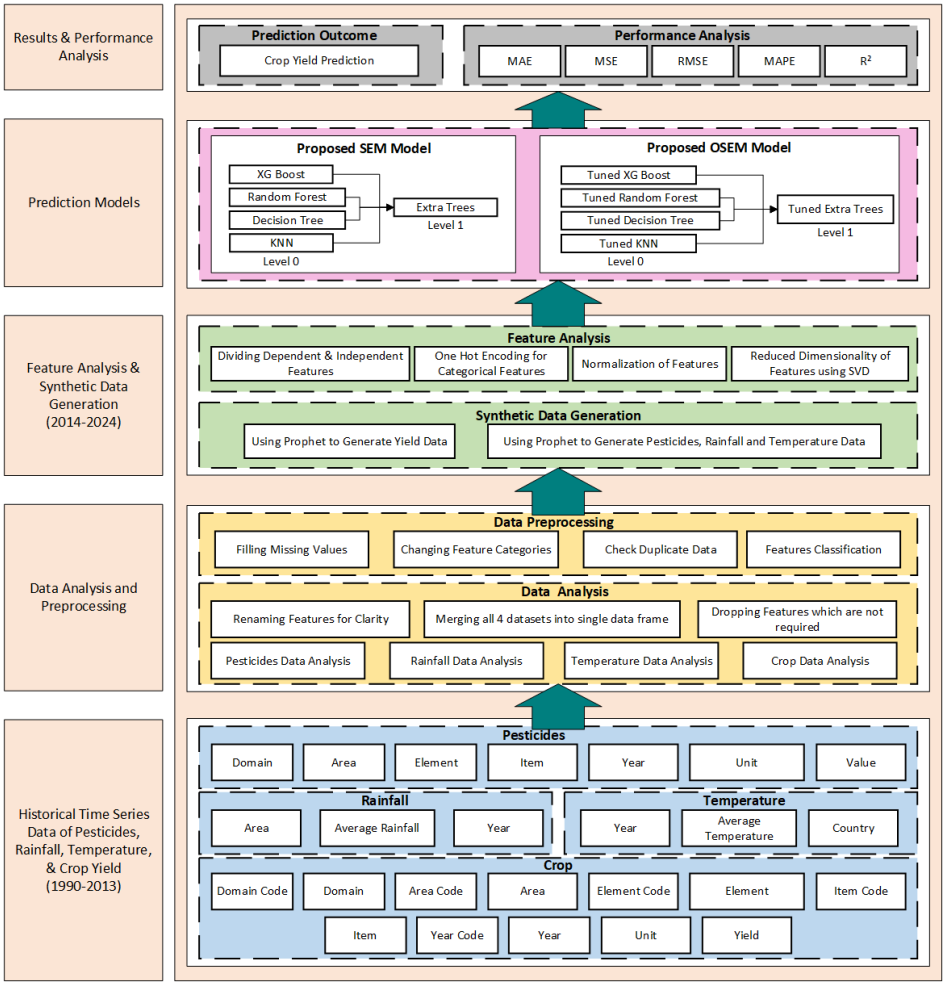


FIGURE 3. Color-coded five layered architecture of proposed crop yield prediction framework.

present in subsection IV-B. The subsection IV-C explains about the feature engineering process, while subsection IV-D summarizes the details of the synthetic data used in this study. The validation of prophet model is present in subsection IV-E and finally, exploratory data analysis is presented in subsection IV-F.

A. DATA COLLECTION

This study uses four datasets that are collected from two different sources. Pesticides tonnes and crop yield datasets are gathered from Food and Agriculture Organization (FAO) [45], while climatic data containing average temperature and rainfall datasets are extracted using the World Data Bank [46]. Table 3 provides a detail summary of all four datasets collected. Pesticides data contain information about different pesticides being used globally from 1990 to 2016. Temperature data consists of different average temperature values from year 1743 to 2013. Rainfall has data of average rainfall per year during 1985 to 2017. At last, the yield data consists of different crops produced globally with their yield value from 1961 to 2016.

TABLE 3. Summary of all 4 datasets collected.

Dataset	Key Columns	Data Range	Description	Total Rows
Rainfall	Area, Year, average_rain _fall_mm_ per_year	1985 to 2017	Annual average rainfall (mm) of different countries.	6727
Pesticides	Domain, Area, Item, Year, Unit, Value	1990 to 2016	Pesticides data in tonnes of active ingredients used.	4349
Temper-ature	year, country, avg_temp	1743 to 2013	Average tempera-ture per year °C around world.	71311
Yield	Domain, Area, Year, Item, Value	1961 to 2016	Top 10 crop yield (hg/ha) data of various countries.	56717

B. DATA PREPROCESSING

At first, all four different datasets collected are cleaned, which includes dropping of unnecessary features, that have no impact on our model prediction. Next, the features are renamed in such a way that they are convenient to use and gives some meaning to the model. After that, all four datasets are merged into a single data frame on the basis of 'year' and 'area' features. An extremely small amount of

TABLE 4. Summary of actual numerical data.

Feature	Data Type	Min.	Max.	Mean	Std.
Year	int64	1990	2013	2001.54	7.0519
Average_ Rainfall (mm)	int64	51	3240	1149.05	709.81
Average_Temperature (°C)	float64	1.3	30.65	20.54	6.3120
Yield (hg/ha)	int64	50	501412	77053.33	84956.61
Pesticides (tonnes)	float64	0.04	4677.78	1707.91	1195.78

TABLE 5. Summary of actual categorical data.

Feature	Unique Values	Most Frequent Value	Freq.	Least Frequent Value	Freq.
Area	100	India	4048	Montenegro	24
Item	10	Potatoes	4276	Plantains and others	556

data contains null values, so null value rows are dropped. The datatype of the average rainfall feature is object that is not desired, so its datatype is converted to float. Finally, the numerical and categorical columns are filtered out, which will be further used in the coming subsections of feature scaling and exploratory analysis. A total of 28,242 instances of data from year 1990 to 2013 were left after preprocessing. Table 4 shows the details of all the numerical features, while the details of categorical features are shown in Table 5.

Algorithm 2 Optimized Stacking Ensemble Model (OSEM)

Input: Training data $X_{\text{train}}, Y_{\text{train}}$
Output: Final prediction $\hat{y}_{\text{final}}$
Initialize Base Learners $f_1, f_2, \ldots, f_n$ (e.g., XGBoost, Random Forest, Decision Tree, KNN)
for each base learner f_i **do**
  Perform hyperparameter tuning: $\theta_i^* = \arg\min_{\theta_i} \mathcal{L}(f_i(X_{\text{train}}, \theta_i), Y_{\text{train}})$
  Train base model f_i on $X_{\text{train}}, Y_{\text{train}}$ with θ_i^*
  Predict crop yield $\hat{y}_i = f_i(X_{\text{train}}, \theta_i^*)$
end for
Stack predictions from all base models: $\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n$
Apply Singular Value Decomposition (SVD) for dimensionality reduction: $\hat{Y} = \text{SVD}(\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n)$
Initialize meta-learner M (Extra Trees Regressor)
Train meta-learner M on the reduced predictions: $\hat{Y}$
Final prediction $\hat{y}_{\text{final}} = M(\hat{Y})$
Return: $\hat{y}_{\text{final}}$

C. FEATURE ENGINEERING

In this, we first divided the input features against the target feature which is yield. Since the area and item features are categorical features, so we used One Hot Encoding to convert them into numerical features. To further normalize the features, we used the Standard Scalar method, by rescaling them to an average of 0 and standard deviation of 1. Then we applied SVD, which is a linear algebra method that

TABLE 6. Summary of synthetic numerical data.

Feature	Data Type	Min.	Max.	Mean	Std.
Year	int64	2014	2024	2019.0	3.16
Average_ Rainfall (mm)	int64	55	4540	1819.43	744.33
Average_Temperature (°C)	float64	3.50	32.44	22.65	7.61
Yield (hg/ha)	int64	43	594440	32087.45	41851.58
Pesticides (tonnes)	float64	0.25	5295.89	2210.71	2005.81

TABLE 7. Summary of synthetic categorical data.

Feature	Unique Values	Most Frequent Value	Freq.	Least Frequent Value	Freq.
Area	100	Australia	110	Kenya	20
Item	10	Potatoes	1034	Yams	220

decomposes a matrix into three matrices retaining only the most significant values among them, resulting in a reduced dimensionality of features [47]. We applied SVD to the features because we want to reduce the noise in the features, capture the important patterns and relevant variables, and enhance the overall efficiency of the system.

D. SYNTHETIC DATA GENERATION

Synthetic data generation is a method used where we want to generate artificial or simulated data based on previous historical datasets, by generating statistical properties and relationships of the actual data [48]. In this study, we have used Prophet model [49], which predicted future trends based on historical data of crop yield from 1990 to 2013. Since the prophet model can only generate one feature at a time, we used it accordingly and generated synthetic data for average rainfall, average temperature, pesticides tone, and crop yield based on area, year and crop name. Table 6 shows the detail of numerical data and Table 7 highlights the detail of categorical data generated using prophet model. In total, we have generated 6,435 samples for crop yield and climatic trends for year 2014-2024. Figure 4 shows the comparison of synthetic data with the actual data. These synthetic data also help in the validation and testing process, enabling the proposed models to give predictions on unseen data. They also provide a broader range of scenarios that may not be present in the available historical dataset.

E. PROPHET MODEL VALIDATION

To assess the reliability and representativeness of the synthetic data generated using the Prophet model, we performed a comparative analysis across three time horizons: 1990-2001 (Actual Early, 12 years), 2002-2013 (Actual Late, 12 years), and 2014-2024 (Synthetic, 11 years). Table 8 presents the mean and standard deviation of four key features (rainfall, temperature, pesticide usage, and crop yield) across these periods. Rainfall and temperature remain highly stable, with variations in mean and spread within $\pm 3\%$, demonstrating that Prophet maintains seasonality and temporal consistency.

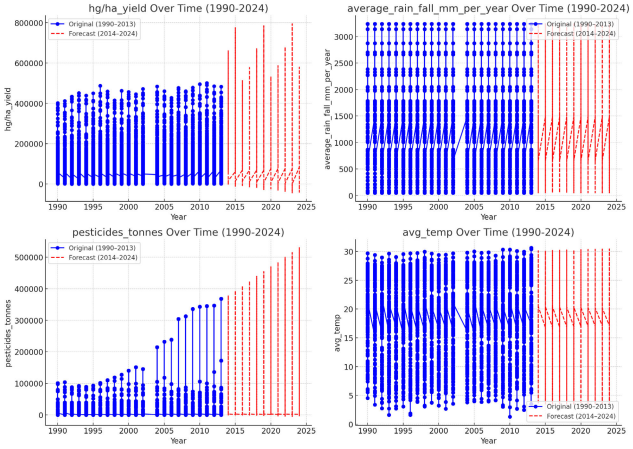


FIGURE 4. Comparison of the features generated through prophet model with actual features data.

TABLE 8. Mean and standard deviation across time horizons.

Horizon / Metric	Rainfall	Temp.	Yield	Pesticides
Mean (1990-2001)	1150.80	20.45	70337.14	29969.75
Mean (2002-2013)	1147.19	20.64	84205.58	44645.51
Mean (2014-2024)	1149.03	20.12	90874.57	54071.03
Std. (1990-2001)	710.04	6.29	77190.76	33321.72
Std. (2002-2013)	709.59	6.34	91977.88	78296.17
Std. (2014-2024)	707.00	6.61	103518.27	95580.79

Crop yield increases progressively across horizons, aligning with long-term agricultural trends. Pesticide usage shows a sharp increase in the late actual period (49%) and then a moderated growth into the synthetic horizon (21%). This moderation can be attributed to both the structural smoothing nature of Prophet and the slightly shorter duration (11 years) of the synthetic period, which naturally reduces cumulative variability compared to the 12-year actual periods.

These observations are further visualized in Figure 5, which shows the percentage change in mean values across the time horizons, complementing the tabulated summary. Together, these results confirm the reliability of the generated synthetic data and support the use of Prophet for producing realistic agricultural time series with consistent trend behavior.

To ensure the synthetic data generated by Prophet does not replicate patterns from the training set, we conducted statistical similarity tests between the actual data (1990-2013) and the synthetic data (2014-2024) across key features. Kolmogorov-Smirnov (KS) [50] test showed statistically significant differences in distributions for all features ($p < 0.001$), confirming that the synthetic values are drawn from a distinct distribution. Additionally, Pearson correlation coefficients [51] were near zero for temperature (-0.0476), rainfall (-0.0035), pesticides ($+0.0320$), and yield (-0.0073), indicating minimal linear similarity. These results, summarized in Table 9, validate that the synthetic data does not overlap structurally with the training data, reducing

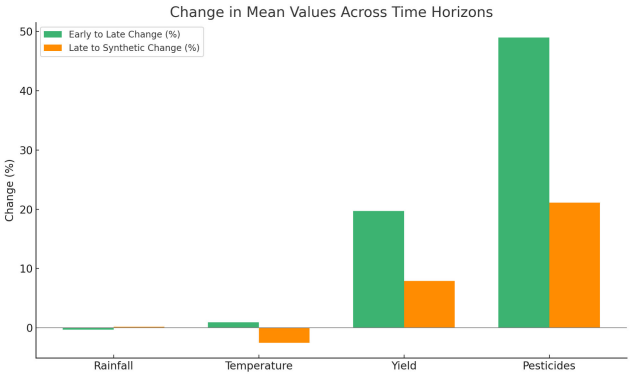


FIGURE 5. Percentage change in mean across 3 horizons.

TABLE 9. Similarity tests between actual and synthetic data.

Feature	KS Statistic	KS p-value	Pearson Correlation
Temperature	0.0740	0.0000	–0.0476
Rainfall	0.0840	0.0000	–0.0035
Pesticides	0.2743	0.0000	+0.0320
Yield	0.0863	0.0000	–0.0073

the risk of inflated performance. Importantly, we clarify that synthetic data was used strictly for validation and testing purposes, while the model was trained only on actual historical data from 1990 to 2013.

F. EXPLORATORY DATA ANALYSIS (EDA)

This subsection introduces a detailed exploratory data analysis of the dataset used in this study. It is an important process in data analysis that involves visually and statistically summarizing the key characteristics of the dataset. We integrated synthetic data with actual data to find the brief analysis of different variables over time and how they relate to each other for their use in crop yield forecasting. Our goal behind EDA is to look for any underlying patterns, find any anomalies, and to identify relationships between features using EDA.

Figure 6 shows the line plot of crop yield against different variables, including pesticides tonnes, average rainfall, and average temperature. We can see in the figure how these three variables could impact the overall yield of the crops. Figure 7 shows different plots giving number of information from the data. The count plot tells the information about the different crops count present inside the data. The Bar plot and Box plot provide information on the top 10 crops and countries in terms of yield and average rainfall, respectively. Finally, the heatmap shows the magnitude of numerical features present inside the data and how they are distributed against each other.

V. EXPERIMENTAL SETUP AND RESULTS

This section is all about the details related to the experiments performed and the results achieved afterward. Additionally, proposed models are evaluated with the best known machine

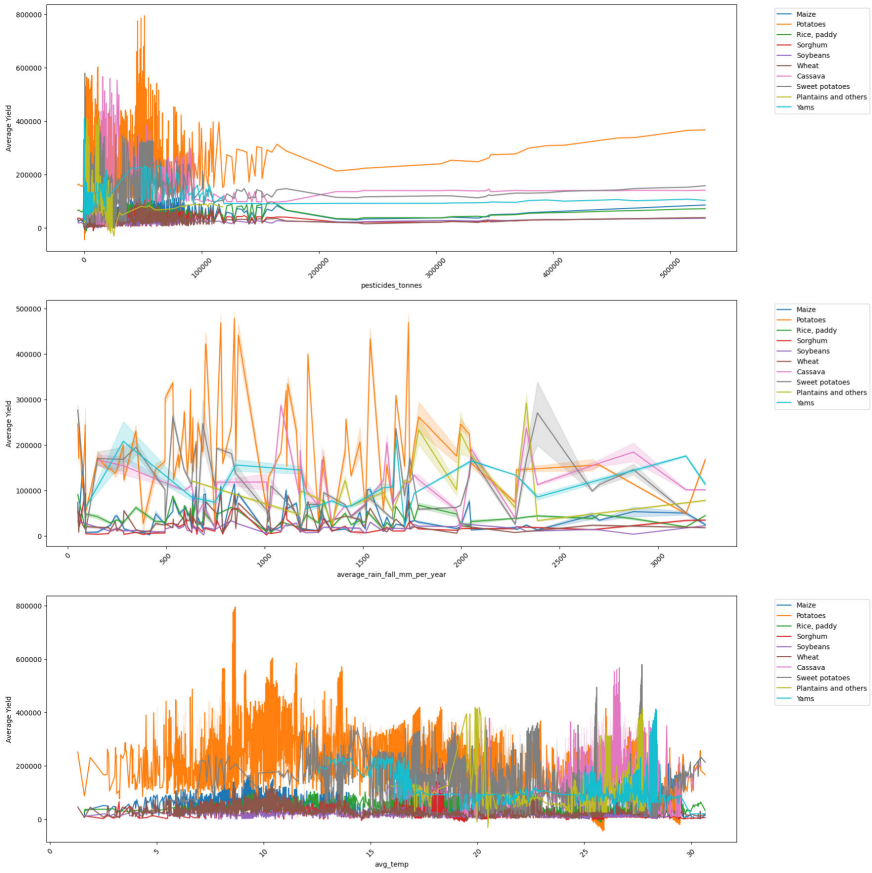


FIGURE 6. Lineplot of crop yield against pesticides, rainfall & temperature.

learning algorithms. Subsection V-A provides the details of all the experimental setup, followed by Subsection V-B, which describes all the performance metrics used to evaluate the results. Subsection V-C outlines all the results produced, including evaluations of SEM, OSEM, region-specific outcomes, K-fold cross-validation, and comparisons with Transformers result. Finally, Subsection V-D explains different convergence and stability tests to measure the reliability of results.

A. EXPERIMENTAL SETUP

This section provides details about all the hardware, software, and other dependencies that were utilized to conduct smooth experiments. The details of all the requirements and their description is given in Table 10. We relied on high performance computing resources, including a multi-core processor and graphical processing units (GPUs). We used Python language on all the experiments performed. Pycharm was used to perform all the data processing and results generation while Jupyter was specifically used for exploratory data analysis and graphical representations of results.

B. PERFORMANCE METRICS

To evaluate and find the performance of our proposed models, we have relied on five efficient and well-known metrics:

TABLE 10. Environmental setup for the proposed SEM & OSEM models.

System Specifications	Description
Operating System	Windows 11 Home
Processor	Intel (R) Xeon(R) Gold 6252 CPU @ 2.10GHz (2 Processors)
Memory	DDR4 704GB
GPU	(2 GPUs) NVIDIA GetForce RTX 4080 16GBytes
IDE	Pycharm Professional 2024.3.4
Python Version	3.10.14
Libraries for Data Processing	pandas, numpy, sklearn.preprocessing
Libraries for Model Selection & Evaluation	sklearn.model_selection, sklearn.metrics, sklearn.pipeline
Libraries for Regression Models	sklearn.ensemble, sklearn.tree, sklearn.neighbors, xgboost, catboost
Libraries for Dimensionality Reduction & Graph Plotting	sklearn.decomposition, matplotlib.pyplot, seaborn

1) MEAN SQUARE ERROR (MSE)

MSE is used to measure the mean square of the differences between the predicted yield and the actual yield. It is commonly used in regression tasks to determine the accuracy of the model [52]. Equation 1 is used to calculate MSE.

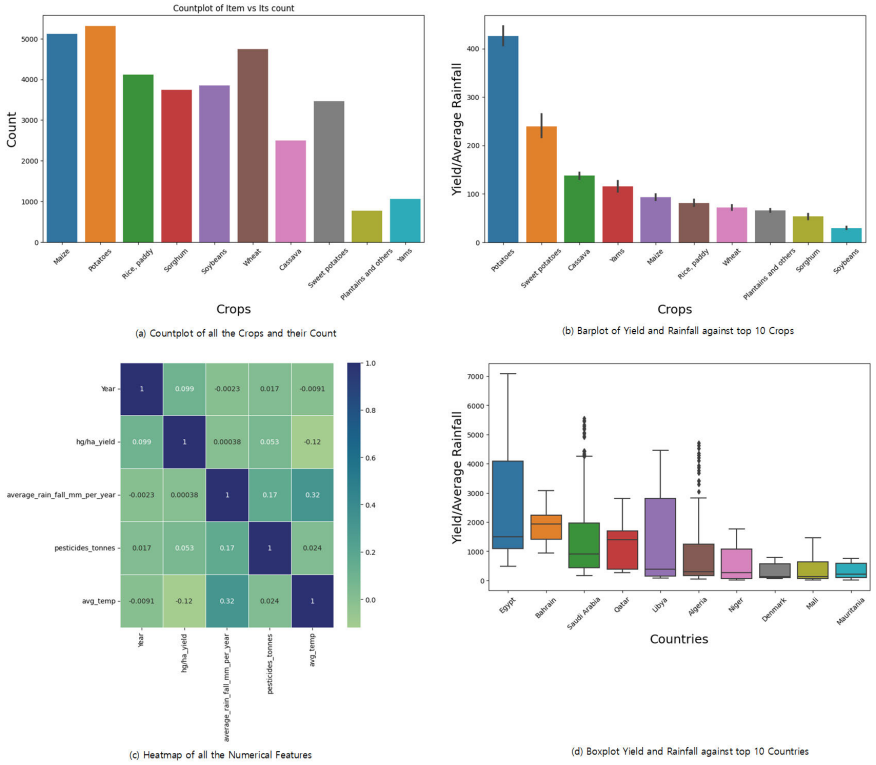


FIGURE 7. Analysis of different features using (a) Countplot, (b) Barplot, (c) Heatmap, and (d) Boxplot.

TABLE 11. Performance evaluation of proposed SEM against 7 machine learning algorithms.

Models	Actual Dataset (1990-2013)					Actual + Synthetic Dataset (1990-2024)				
	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2
Gradient Boost	205369.34	453.17	2.746	113.09	0.739	161483.43	401.85	2.471	60.95	0.777
Ada Boost	621877.51	788.59	6.971	491.81	0.212	334788.78	578.60	5.786	206.45	0.538
XG Boost	28682.87	169.36	0.986	32.63	0.963	20446.61	142.99	0.823	20.57	0.971
Cat Boost	30438.02	174.46	1.081	43.54	0.961	23994.39	154.90	0.965	26.73	0.966
Random Forest	9263.97	96.24	0.391	9.23	0.982	8908.79	94.38	0.344	7.33	0.987
Decision Tree	16047.19	126.67	0.404	9.56	0.979	14463.54	120.26	0.353	7.45	0.980
K-Nearest Neighbor	10520.22	102.56	0.436	9.42	0.981	11697.78	108.15	0.437	8.47	0.983
SEM	**8832.36**	**93.98**	**0.363**	**8.71**	**0.985**	**2584.31**	**50.83**	**0.278**	**6.78**	**0.990**

2) ROOT MEAN SQUARE ERROR (RMSE)

RMSE is the under root of MSE. It gives magnitude of the error in the same scale as actual data, with lower value means high performance [53]. It is calculated using equation 2.

3) MEAN ABSOLUTE ERROR (MAE)

MAE examines the mean of the absolute differences among the predicted and actual yield. It provides simple mean of the absolute errors, making it easier to understand the evaluation of the model [54]. To calculate MAE, equation 3 is used.

4) MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

MAPE calculates the percentage of mean absolute error among the predicted and actual output. We can get a percentage for comparing the performance across different datasets using MAPE [54]. Equation 4 is used to calculate MAPE.

5) COEFFICIENT OF DETERMINATION (R^2)

R^2 calculates the proportion of the variance of the dependent feature from the independent feature [55]. It ranges from 0 to 1, with values closed to 1 indicates that the model shows a greater part of variance. Equation 5 is used to find

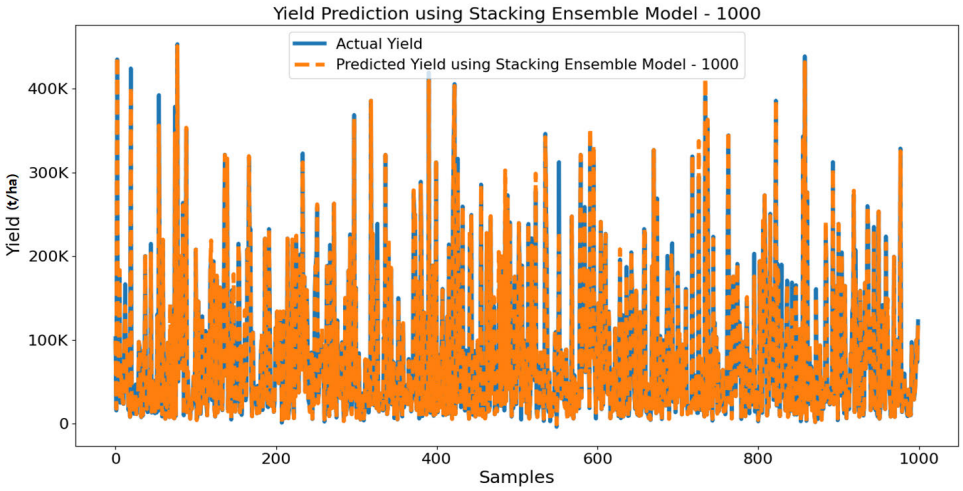


FIGURE 8. Comparison of actual vs. predicted yield across 1000 samples using SEM with actual+synthetic data.

R^2.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3}$$

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{4}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{5}$$

where:

- y_i is the actual value for the i-th data point.
- $\hat{y}_i$ is the predicted value.
- $\bar{y}$ is the mean of the actual values.
- n is the number of observations.

C. PREDICTION RESULTS

This section outlines the detailed experiments conducted on the proposed models using the actual dataset and the integration of actual with synthetic data. We evaluated SEM and the OSEM models on given datasets and evaluation results are compared with previously used and popular regression-based machine learning algorithms.

1) EVALUATION USING SEM

The Stacking Ensemble Model (SEM) was evaluated using two distinct experimental settings to assess its robustness and adaptability. In the first experiment, the SEM was trained and tested entirely on actual crop yield data collected from 1990 to 2013. A total of 22,593 samples were used for training and 5,649 for testing, using an 80:20 random split. In the second experiment, a more realistic forecasting scenario was created by training SEM on historical data and testing it solely on synthetic data. Specifically, 28,242 samples of actual data were used for training, while 6,435 synthetic samples were used exclusively for testing. This split was designed to evaluate the model's ability to generalize beyond observed historical trends and capture future crop yield variations.

TABLE 12. Results of SEM using synthetic data.

	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2
Training	2475.25	49.75	0.185	4.28	0.996
Validation	8939.40	94.54	0.416	7.84	0.987
Testing	2584.31	50.83	0.278	6.78	0.990

Table 11 presents the performance of SEM in both experimental setups across five evaluation metrics: R^2, MAE, MAPE, MSE, and RMSE. The yield values, originally in hectograms per hectare (hg/ha), were converted to tons per hectare (t/ha) to ensure interpretable error magnitudes. In both scenarios, SEM outperforms all baseline regression models. When tested on actual data, SEM achieved an R^2 of 0.985 and a MAPE of 8.71%. Notably, performance improved when SEM was tested on the synthetic dataset, reaching an R^2 of 0.990 and a reduced MAPE of 6.78%, indicating superior generalization to unseen future-like patterns. The reductions in MAE, MSE, and RMSE further reinforce the enhanced accuracy of SEM in the second setting.

To further assess the stability of SEM on unseen synthetic data, an additional training-validation-testing split of 70%-15%-15% was conducted exclusively on the combined dataset. Table 12 shows consistent performance across all phases, confirming that SEM generalizes well and avoids

TABLE 13. Performance evaluation of proposed OSEM against 7 optimized machine learning algorithms.

Models	Actual Dataset (1990-2013)					Actual + Synthetic Dataset (1990-2024)				
	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2
Optimized Gradient Boost	56749.78	238.22	1.443	54.17	0.928	40607.48	201.51	1.232	33.09	0.944
Optimized Ada Boost	395442.27	628.84	4.731	270.78	0.499	343582.56	586.15	4.718	212.12	0.526
Optimized XG Boost	61015.31	247.01	1.484	56.70	0.922	43049.48	207.48	1.269	34.25	0.940
Optimized Cat Boost	80805.69	284.26	1.817	80.23	0.897	58693.48	242.26	1.552	41.87	0.919
Optimized Random Forest	22244.45	149.14	0.789	32.73	0.971	21569.78	146.86	0.719	21.19	0.970
Optimized Decision Tree	32863.96	181.28	0.893	31.15	0.958	30739.88	175.32	0.792	23.86	0.957
Optimized K-Nearest Neighbor	8570.79	92.57	0.375	8.69	0.983	7901.51	88.89	0.345	6.77	0.989
Optimized SEM	**8555.15**	**92.49**	**0.367**	**8.20**	**0.989**	**2475.25**	**49.75**	**0.185**	**4.28**	**0.996**

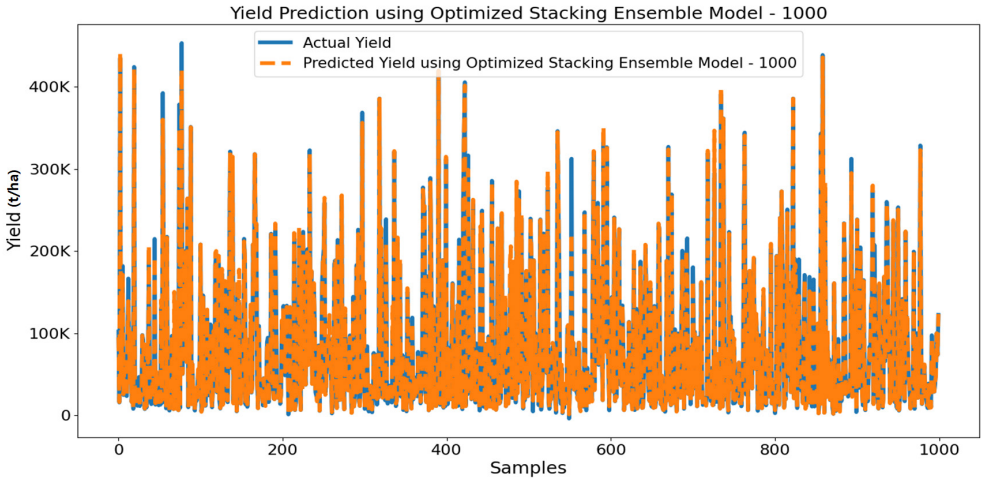


FIGURE 9. Comparison of actual vs. predicted yield across 1000 samples using OSEM with actual+synthetic data.

overfitting. Additionally, Figure 8 illustrates the correlation between actual and predicted crop yields across the first 1,000 test samples. The close alignment of predictions with actual values highlights the reliability and precision of SEM in forecasting yield trends.

2) EVALUATION USING OSEM

Following the SEM framework, we further proposed the Optimized Stacking Ensemble Model (OSEM) to enhance prediction accuracy and generalization. The OSEM architecture incorporates two key improvements: dimensionality reduction and hyperparameter tuning. First, Singular Value Decomposition (SVD) was applied to extract 80% of the most informative features, effectively reducing noise and computational complexity. Next, GridSearchCV was employed to tune hyperparameters for each base learner and the meta-learner, using exhaustive search with cross-validation to identify the optimal model configurations.

OSEM was evaluated using the same experimental setups and data splits previously described for SEM. Table 13 summarizes OSEM's performance against other traditional machine learning models, all of which were optimized using the same GridSearchCV strategy. On the synthetic test dataset, OSEM achieved an R^2 of 0.996, MAE of 0.185 t/ha, and MAPE of 4.28%, outperforming all baseline models and confirming its superior generalization capacity. When evaluated on actual test data, OSEM also achieved a strong R^2 of 0.989 and MAPE of 8.20%. Figure 9 illustrates the comparison of actual and predicted yields across 1,000 test samples, further validating the model's predictive strength and reduced error margins compared to SEM.

In addition, OSEM was subjected to a train-validation-test evaluation on the integrated dataset, with 70% of the data used for training and 15% each for validation and testing. Table 14 presents the results across these three phases, demonstrating that OSEM maintains high performance and consistency across splits, with no indication of overfitting. These results confirm the effectiveness of integrating dimensionality reduction and model tuning into the ensemble framework for robust and scalable crop yield forecasting.

TABLE 14. Results of OSEM using synthetic data.

	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2
Training	1753.13	41.87	0.157	3.55	0.998
Validation	8461.65	91.98	0.360	7.39	0.988
Testing	2475.25	49.75	0.185	4.28	0.996

TABLE 15. Results of OSEM model for top 5 regions.

Region	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2
India	5476.76	74.01	0.100	1.33	0.995
Brazil	210.72	14.52	0.069	1.60	0.997
Mexico	2689.00	51.86	0.306	4.10	0.996
Pakistan	2000.28	44.72	0.159	3.03	0.993
Japan	3388.36	58.21	0.268	1.92	0.997

3) EVALUATION USING OSEM BY REGION

To investigate the geographical adaptability of our proposed model, we performed region-specific analysis on the top five most represented regions: India, Brazil, Mexico, Pakistan, and Japan. As shown in Table 15, the OSEM model consistently achieved high R^2 scores (≥ 0.993) across all regions, with the lowest MAPE of 1.33% in India and highest MAE of 0.306 t/ha in Mexico. This region-wise analysis was conducted to assess the robustness of the model across heterogeneous agricultural and climatic conditions in different regions around the world.

4) K-FOLD CROSS VALIDATION (CV)

K-Fold cross-validation is a key method for assessing a model's robustness and generalization. In this study, we employed a 10-fold cross-validation approach tailored to our hybrid dataset setup. For each fold, we randomly selected 9 subsets from the actual historical data (1990-2013) for training, and 1 subset from the synthetic data (2014-2024) for testing. This configuration ensures strict temporal separation between training and testing while simulating how well the model generalizes to unseen future-like scenarios generated via the Prophet model.

Table 16 shows the evaluation results across all folds. Notably, several R^2 values of OSEM exceed 0.99, which demonstrates that the model does not overfit to the training data but rather generalizes effectively to the synthetic data. The low variance in results across folds further confirms the stability and reliability of both SEM and OSEM under diverse data partitions.

5) EVALUATION USING TRANSFORMER-BASED MODELS

To further validate the robustness of our proposed models, we conducted experiments using transformer-based models that are designed for tabular data [56], [57]. Specifically, we evaluated the performance of the TabNet model across three different training durations (100, 150, and 200 epochs) and observed progressive improvements. Moreover, we developed an additional stacking ensemble configuration that uses TabNet and TabPFN as base learners and employs Extra Trees Regressor as the meta-learner. The purpose was to benchmark our optimized ensemble (OSEM) against state-of-the-art transformer architectures. As presented in Table 17, although transformer-based models showed decent performance, particularly in the ensemble form, our OSEM model consistently outperformed them across all metrics including R^2, MAE, and MAPE, confirming its superior predictive capacity and generalizability.

TABLE 16. 10-Fold cross validation results of SEM and OSEM.

Fold	SEM		OSEM	
	MAE (t/ha)	R^2	MAE (t/ha)	R^2
1	0.317	0.989	0.193	0.993
2	0.294	0.989	0.189	0.990
3	0.326	0.989	0.186	0.990
4	0.297	0.991	0.203	0.992
5	0.330	0.987	0.229	0.994
6	0.335	0.990	0.208	0.996
7	0.305	0.991	0.193	0.993
8	0.322	0.990	0.217	0.994
9	0.330	0.988	0.210	0.992
10	0.304	0.987	0.222	0.990

TABLE 17. Performance of transformer-based models on crop yield Prediction.

Model	MSE (t/ha)	RMSE (t/ha)	MAE (t/ha)	MAPE (%)	R^2
TabNet (100 epochs)	160008.63	400.01	2.609	149.91	0.800
TabNet (150 epochs)	103902.01	322.34	2.084	111.12	0.870
TabNet (200 epochs)	82416.83	287.08	1.855	104.13	0.897
TabNet+TabPFN +ETR	29442.88	171.59	0.950	22.51	0.959
Proposed OSEM	2475.25	49.75	0.185	4.28	0.996

D. CONVERGENCE AND STABILITY ANALYSIS

To assess the consistency and generalization ability of our proposed SEM and OSEM models, we conducted a detailed convergence and stability analysis using 10-fold cross-validation. In addition to SEM and OSEM, we also evaluated optimized baseline machine learning models, including XGBoost, Random Forest, Decision Tree, and KNN regressors. For each model, we computed the mean and standard deviation of MAE and R^2 scores across all folds. As shown in Table 18, OSEM achieved the lowest MAE (0.2071 ± 0.0148) and highest R^2 (0.9924 ± 0.0021), indicating strong predictive accuracy and stability. Complementary visualizations are presented in Fig. 10

TABLE 18. Convergence stability analysis across 10 folds.

Model	MAE (Mean ± Std.)	R^2 (Mean ± Std.)
Decision Tree	0.8281 ± 0.0337	0.9518 ± 0.0020
KNN	0.3694 ± 0.0254	0.9845 ± 0.0020
XGBoost	1.3118 ± 0.0557	0.9373 ± 0.0026
Random Forest	0.8280 ± 0.0260	0.9689 ± 0.0017
SEM	0.3121 ± 0.0141	0.9891 ± 0.0013
OSEM	0.2071 ± 0.0148	0.9924 ± 0.0021

(boxplot of MAE) and Fig. 11 (line plot of R^2), which illustrate minimal performance variation for OSEM across folds compared to other models.

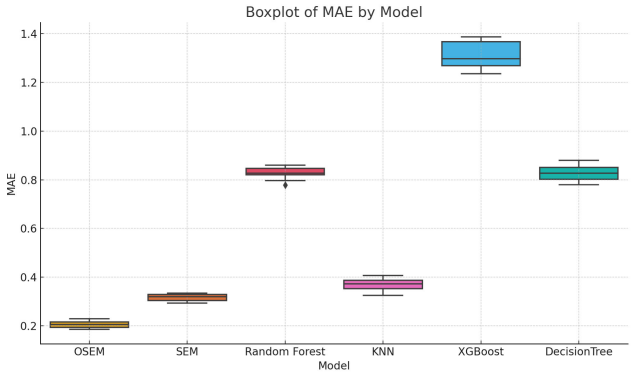


FIGURE 10. Box plot of MAE values across 10-folds for all models.

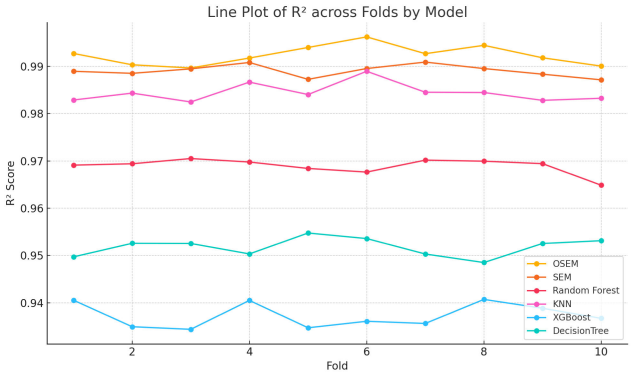


FIGURE 11. Line plot of R^2 scores across 10-folds for all models.

To further validate statistical significance, we conducted both Wilcoxon signed-rank [58] and Friedman tests [59] based on MAE values across 10-folds. The Friedman test yielded a highly significant result ($\chi^2(5) = 48.63$, $p < 0.00001$), confirming meaningful performance differences among the models. Average ranks derived from the Friedman test (Table 19) placed OSEM as the top-performing model with a perfect rank of 1.00. Additionally, the Wilcoxon signed-rank test between SEM and OSEM (Table 20) demonstrated a statistically significant advantage of OSEM over SEM ($p = 0.00195$). These statistics were computed

TABLE 19. Friedman test: Average mae-based rankings across 10 folds.

Model	Average Rank
OSEM	1.00
SEM	2.00
KNN	3.00
Decision Tree	4.40
Random Forest	4.60
XGBoost	6.00

TABLE 20. Wilcoxon signed-rank test between SEM and OSEM models (MAE).

Model 1	Model 2	Statistic	p-value
SEM	OSEM	0.0	0.00195

using the standard Friedman test formula (Eq. 6) and Wilcoxon rank formula (Eq. 7) as outlined in the statistical formulation subsection. Collectively, these results reinforce the effectiveness and consistency of our optimized ensemble model.

$$\chi_F^2 = \frac{12n}{k(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1) \tag{6}$$

$$W = \sum_{i=1}^{n} R_i \cdot \mathrm{sgn}(D_i) \tag{7}$$

VI. DISCUSSION

This study addresses the challenge of accurate and efficient crop yield prediction using the proposed Stacking Ensemble Model (SEM) and its optimized variant (OSEM). Both models incorporate four machine learning regressors including XGBoost, Random Forest, Decision Tree, and KNN as base learners, with Extra Trees serving as the meta-learner. To enhance generalization, we integrated synthetic data generated using the Prophet model, simulating future climatic and agricultural conditions such as pesticide usage and yield variability. This hybrid approach enabled the models to learn from both real-world and synthetic edge-case scenarios, improving their robustness under varying conditions.

The results demonstrate strong predictive performance. As shown in Figure 12, both SEM and OSEM maintain close alignment between predicted and actual crop yields, with predictions clustering along the ideal diagonal. OSEM achieves a lower Mean Absolute Error (MAE) of 0.185 t/ha (185 kg/ha average yield deviation) compared to SEM's 0.278 t/ha (278 kg/ha), reflecting improved accuracy for yield-sensitive decisions. The R^2 score of 0.996 indicates OSEM explains 99.6% of the variance in crop yield. Improvements in RMSE and MAPE further confirm OSEM's superior forecasting reliability, which is essential for practical applications such as fertilizer planning, irrigation scheduling, and agricultural risk management.

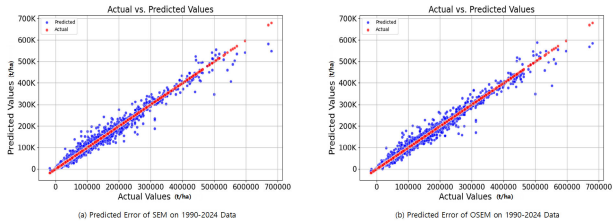


FIGURE 12. Actual vs. Predicted yield of (a) SEM and (b) OSEM on actual+synthetic dataset.

TABLE 21. Comparison of SEM & OSEM with prior studies.

Studies	Methods Used	MAE	R^2
Najjar et al. [22]	LSTM	0.35	0.76
Zhang et al. [23]	LSTM-MH-SA	-	0.81
Oikonomidis et al. [25]	CNN-DNN	0.199	-
Yadav et al. [28]	BR, KNN, DT, GB, RF, LR	0.10	0.985
Sharma et al. [29]	RF, DT, XGB	1.97	0.989
Shafi et al. [31]	RF, XGB, LASSO	21.72	0.93
Cedric et al. [34]	Multivariate LR, DT, KNN	0.16	0.953
Proposed SEM	**XGB, RF, DT, KNN**	0.27	**0.990**
Proposed OSEM	**+ ETR**	**0.18**	**0.996**

Furthermore, Table 21 provides a comparative evaluation against recent state-of-the-art (SOTA) models. Our results show that OSEM consistently outperforms traditional ML and DL approaches, particularly under future-like synthetic test conditions. This reflects the robustness and adaptability of model in real-world agricultural forecasting.

In addition, we evaluated TabNet, a transformer-based deep learning model for tabular data, both as a standalone model and within an ensemble with TabPFN and Extra Trees. While TabNet alone achieved an R^2 of 0.897 with MAE = 1.855 t/ha, the stacked TabNet + TabPFN + ETR ensemble improved results to R^2 = 0.959 and MAE = 0.953 t/ha. However, these values still fell short of OSEM, reinforcing the effectiveness of our optimized stacking strategy, which leverages both model diversity and hyperparameter tuning for superior generalization.

To further assess model reliability, we conducted a convergence and stability analysis using 10-fold cross-validation. OSEM not only achieved higher average performance across folds but also demonstrated greater consistency with lower variation in prediction errors. This indicates better generalization and stability under different data splits. Additionally, Convergence analysis and Statistical tests including Wilcoxon signed-rank and Friedman, confirmed the significance of these improvements, highlighting the robustness and dependable behavior of OSEM, especially critical for deployment in operational agricultural decision-making systems.

TABLE 22. Performance impact of OSEM components.

Model Variant	MAE (t/ha)	RMSE (t/ha)	R^2
Base SEM	0.278	50.83	0.990
SEM + SVD	0.253	49.91	0.991
SEM + Hyperparameter Tuning	0.226	47.30	0.993
Full OSEM (SVD+Tuning+Stacking)	**0.185**	**49.75**	**0.996**

A. OSEM ABLATION STUDY

To understand the individual contribution of each optimization component within the OSEM framework, we performed an ablation study evaluating the effects of dimensionality reduction (SVD), hyperparameter tuning, and ensemble stacking. Table 22 presents the performance of four configurations: baseline SEM, SEM with SVD, SEM with tuning, and the full OSEM model. Results show that hyperparameter tuning and stacking provide the most significant performance gains, while dimensionality reduction offers moderate but consistent improvements. These findings validate the architectural choices in OSEM and highlight the value of multi-layer optimization for yield forecasting.

B. LIMITATIONS

While the proposed approach delivers strong performance in crop yield prediction, several limitations exist. First, integrating multiple base learners with SVD-based dimensionality reduction and hyperparameter tuning increases computational cost, especially due to the global scale of the dataset. However, region-specific experiments demonstrated a substantial reduction in compute load, suggesting the model's practicality in localized deployments.

Second, the effectiveness of synthetic data is sensitive to its ability to capture real-world variability; failure to represent extreme conditions or regional nuances may impact accuracy. Additionally, while SVD improves efficiency, it can occasionally eliminate relevant features, potentially reducing performance in certain cases.

Lastly, applying this framework in remote agricultural areas is challenged by limited data availability and computational resources, underscoring the need for adaptable, lightweight solutions for real-world deployment.

VII. CONCLUSION AND FUTURE DIRECTION

In this empirical study, we proposed a Stacking Ensemble Model (SEM) and its optimized variant (OSEM) for accurate and robust crop yield forecasting. Leveraging synthetic data generated through the Prophet model, based on features such as temperature, precipitation, pesticide usage, and historical yield, we demonstrated significant improvements in predictive accuracy and model generalization. The optimized OSEM model achieved an outstanding performance with an R^2 score of 0.996 and a MAE of 0.185 t/ha, suggesting high reliability even under simulated future climate conditions. Beyond performance metrics, we conducted extensive

evaluations to validate model robustness. Convergence and stability analyses across 10-fold cross-validation were carried out using MAE and R^2 scores, supported by statistical significance tests including the Wilcoxon signed-rank and Friedman tests. These analyses confirmed the consistent superiority and reliability of the OSEM model over traditional machine learning baselines.

For future work, we aim to expand our framework by integrating multimodal datasets, including soil quality, satellite imagery, humidity, CO_2 levels, and economic indicators. Additionally, we plan to implement weighted ensemble techniques that dynamically adjust learner contributions while optimizing for computational efficiency. These advancements will further strengthen the applicability of model in agricultural planning, resource optimization, and climate-resilient farming strategies to support global food security.

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